

Assignment 1: Language ID

EE6130: Advanced Topics in Signal Processing

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1 Question

- Try tapping the features from different layers of pre-trained wav2vec2 and report your findings.
- Try HuBERT architecture to extract audio features and compare it with wav2vec2.
- Make the CNN model shown in the demo more complex.
- With the same Dataloading technique as shown in the demo, add two more languages before starting to build the Language Identification model. (choose similar languages, completely different languages)
- Try to improve the accuracy.

Write a two-page report on your findings. Any act of plagiarism will result in zero points for the assignment.

2 Solution - My Work

Google Colab Notebook link: [Link](#)

2.1 Tapping the Features from Different Layers of Pre-trained wav2vec2 and Reporting the Findings

Tapping the features refers to outputs extracted from specific layers of a neural network. This allows us to validate and analyze the feature representation after each layer. To do this, we need to print an instance of the extracted features.

The figure 1 refers to the first record of the subset. This is a tensor having size [516, 768], which implies that there are 516 frames, each embedded as 768 features. This follows the official documentation. We can also observe that the values are less than 1, and some are negative.

```

> print(subset_with_features[0])
print(subset_with_features[0].size())
tensor([[-0.0772, 0.1102, 0.1043, ..., -0.0704, -0.0648, 0.1522],
        [ 0.0648, 0.0273, 0.1109, ..., -0.1113, -0.1181, -0.3096],
        [ 0.2113, 0.0404, 0.1104, ..., 0.0126, 0.0172, 0.0007],
        ...,
        [ 0.0404, 0.0479, 0.0042, ..., 0.1104, 0.1013, 0.1104],
        [ 0.0648, 0.0641, 0.0041, ..., 0.1109, 0.1000, 0.1104],
        [ 0.0100, 0.0104, 0.0073, ..., -0.1101, 0.0000, 0.1019]])
torch.Size([1, 514, 768])

```

Figure 1: An instance of the subset using wav2vec2

2.2 Trying HuBERT architecture to extract audio features and compare it with the wav2vec2

```

[21]: print(subset_with_features2[0])
print(subset_with_features2[0].size())

tensor([[[ 0.1948, 0.4494, 0.1412, ..., -0.3799, -0.0665, 0.0863],
         [ 0.3062, 0.3740, -0.1544, ..., -0.6417, 0.0487, 0.3728],
         [ 0.4065, 0.2213, -0.2973, ..., -0.1695, -0.3489, 0.4849],
         ...,
         [ 0.1643, 0.6335, 0.3182, ..., 0.0820, -0.1277, 0.1822],
         [ 0.1185, 0.6964, 0.4073, ..., 0.1414, -0.1695, 0.1246],
         [ 0.0264, 0.6362, 0.3412, ..., 0.1534, -0.1997, -0.0610]]]])
torch.Size([1, 514, 1024])

```

Figure 2: An instance of the subset using HuBERT

In this case, as we can see in figure 2, we get a tensor of size [514, 1024], which implies that there are 514 frames and 1024 embedded features. This also follows the official documentation.

2.3 Make the CNN more complex

Figure 3 shows the new addition of a convolutional layer to the model. The accuracy has increased from 60.83 % to 83.33 %. The metrics are shown in further subsections.

```

class CNNModel(nn.Module):
    def __init__(self, input_dim, num_classes):
        super(CNNModel, self).__init__()
        self.conv1 = nn.Conv1d(in_channels=input_dim, out_channels=128, kernel_size=3, padding=1)
        self.conv2 = nn.Conv1d(in_channels=128, out_channels=256, kernel_size=3, padding=1)
        self.conv3 = nn.Conv1d(in_channels=256, out_channels=512, kernel_size=3, padding=1)
        self.fc1 = nn.Linear(512 * (input_length // 8), 256) # Adjust the input dimension to the linear layer
        self.fc2 = nn.Linear(256, num_classes)

    def forward(self, x):
        x = F.relu(self.conv1(x))
        x = F.max_pool1d(x, kernel_size=2)
        x = F.relu(self.conv2(x))
        x = F.max_pool1d(x, kernel_size=2)
        x = F.relu(self.conv3(x))
        x = F.max_pool1d(x, kernel_size=2)
        x = x.view(x.size(0), -1)
        x = F.relu(self.fc1(x))
        x = self.fc2(x)
        return x

```

Figure 3: New CNN model

2.4 Adding two more languages to the data loading and building the language identification model

I chose to add German and Polish languages. In the same way as shown in demo pynb was done. As shown in figures below.

```
[43]: # find the language id of de
subset_copy[601]

[43]: {'audio_id': '20131008-0900-PLINARY-13-de_20131008-17:44:02_1',
      'language': 1,
      'audio': {'path': 'train_part_0/20131008-0900-PLINARY-13-de_20131008-17:44:02_1.wav',
                'array': array([ 0.00030518,  0.00021362, -0.0005188 , ..., -0.00085449,
                                0.00244141,  0.00228882])},
      'sampling_rate': 16000,
      'raw_text': 'Dabei kommt es nicht auf ein bestimmtes Alter oder die Hautfarbe an.',
      'normalized_text': 'dabei kommt es nicht auf ein bestimmtes alter oder die hautfarbe an.',
      'gender': 'female',
      'speaker_id': '97016',
      'is_gold_transcript': True,
      'accent': 'None'}
```

Figure 4: 1st row of the subset for german language

```
[44]: # find the language id of pl
subset_copy[801]

[44]: {'audio_id': '20110216-0900-PLINARY-4-pl_20110216-09:57:42_3',
      'language': 4,
      'audio': {'path': 'train_part_0/20110216-0900-PLINARY-4-pl_20110216-09:57:42_3.wav',
                'array': array([-0.04589844,  0.0513916 ,  0.02088057, ...,  0.01092529,
                                0.00880002,  0.01290894])},
      'sampling_rate': 16000,
      'raw_text': 'Bliska perspektywa zakończenia negocjacji oraz stworzona w tej Izbie atmosfera pomogą również samej Chorwacji.',
      'normalized_text': 'bliska perspektywa zakończenia negocjacji oraz stworzona w tej izbie atmosfera pomogą również samej chorwacji.',
      'gender': 'male',
      'speaker_id': '22785',
      'is_gold_transcript': True,
      'accent': 'None'}
```

Figure 5: 1st row of the subset for polish language

German is a part of Germanic language and Polish is part of Salvic language family. The subset I have taken is mixed subset. Using the wav2vec2 for feature extraction as before, and applying the new CNN model. Here we can see that the accuracy as reduced to 40.50 %. This can be less dataset being supplied for training. Making the model more complex may increase the accuracy even more. If similar languages are taken, the model would have learned much better and the accuracy would go up.

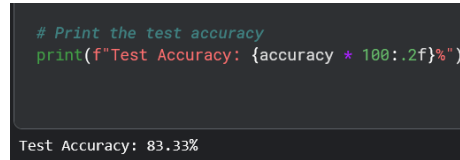
```
# Print the test accuracy
print(f"Test Accuracy: {accuracy * 100:.2f}%")

Test Accuracy: 40.50%
```

Figure 6: Five languages based subset Language Identification Model performance

2.5 Accuracy improvement

Using 3 languages, the new CNN model gave more accuracy than the given demo python notebook.



```
# Print the test accuracy
print(f"Test Accuracy: {accuracy * 100:.2f}%")

Test Accuracy: 83.33%
```

Figure 7: Three languages based subset Language Identification Model performance